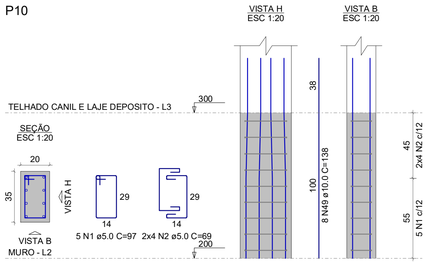
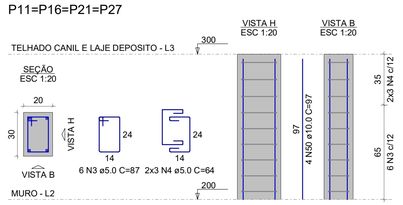
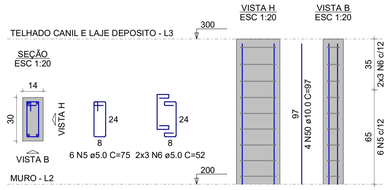
P10



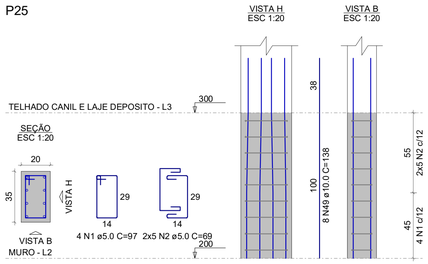
P11=P16=P21=P27



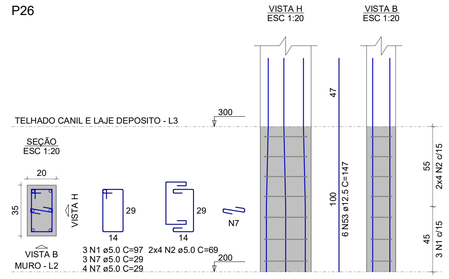
P12=P13=P14=P15=P17=P18=P19=P20=P22=P23=P24= =P28=P29=P30



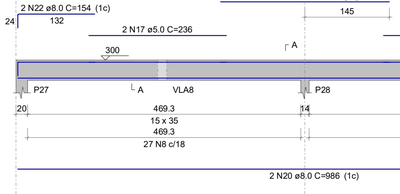
P25



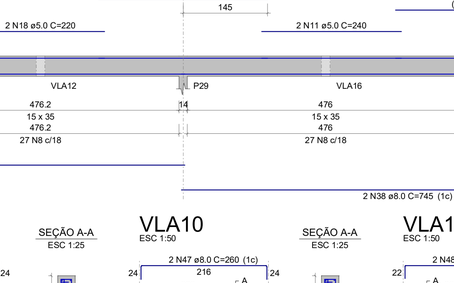
P26



P31

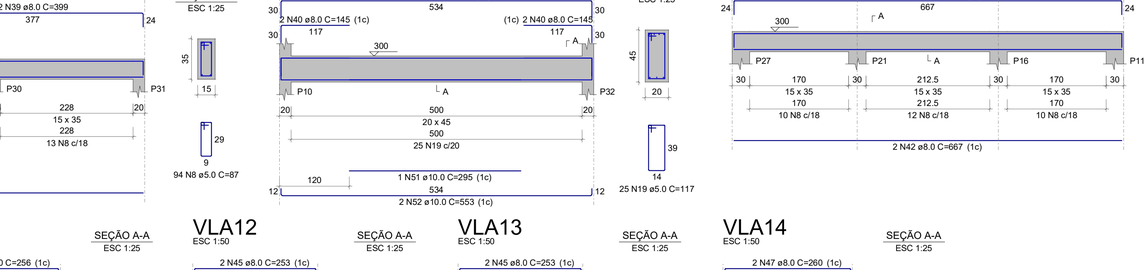


P32



VLA1

ESC 1:50



SEÇÃO A-A
ESC 1:25

VLA2

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA3

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA4

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA5

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA6

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA7

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA8

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA9

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA10

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA11

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA12

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA13

ESC 1:50

SEÇÃO A-A
ESC 1:25

VLA14

ESC 1:50

SEÇÃO A-A
ESC 1:25

RELAÇÃO DO AÇO

P10	4xP11	14xP12
P25	P26	P31
VLA3	VLA4	VLA5
VLA6	VLA7	VLA8
VLA9	VLA10	VLA11
VLA12	VLA13	VLA14

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	16	97	1552
	2	5.0	32	69	2208
	3	5.0	29	87	2523
	4	5.0	32	64	2048
	5	5.0	84	75	6300
	6	5.0	84	52	4368
	7	5.0	14	29	406
	8	5.0	378	87	32886
	9	5.0	2	237	474
	10	5.0	4	220	880
	11	5.0	8	240	1920
	12	5.0	103	85	8755
	13	5.0	2	237	474
	14	5.0	2	232	464
	15	5.0	25	137	3425
	16	5.0	2	235	470
	17	5.0	2	236	472
	18	5.0	2	220	440
	19	5.0	25	117	2925
CA50	20	8.0	4	987	3948
	21	8.0	2	742	1484
	22	8.0	4	154	616
	23	8.0	12	290	3480
	24	8.0	2	396	792
	25	8.0	4	987	3948
	26	8.0	4	742	2968
	27	8.0	2	152	304
	28	8.0	2	394	788
	29	8.0	2	159	318
	30	8.0	2	780	1560
	31	8.0	2	396	792
	32	8.0	3	320	960
	33	8.0	3	550	1650
	34	8.0	1	127	127
	35	8.0	3	187	561
	36	8.0	1	121	121
	37	8.0	3	181	543
	38	8.0	2	745	1490
	39	8.0	2	399	798
	40	8.0	4	145	580
	41	8.0	2	590	1180
	42	8.0	2	667	1334
	43	8.0	2	711	1422
	44	8.0	8	209	1672
	45	8.0	8	253	2024
	46	8.0	6	216	1296
	47	8.0	4	260	1040
	48	8.0	2	256	512
	49	10.0	16	138	2208
	50	10.0	76	97	7372
	51	10.0	1	295	295
	52	10.0	2	553	1106
	53	12.5	12	147	1764

RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	PESO + 10% (kg)
CA50	8.0	383.1	166.3
	10.0	109.8	74.5
	12.5	17.6	18.7
CA60	5.0	729.9	123.8
PESO TOTAL (kg)			
CA50	259.4		
CA60	123.8		

Volume de concreto (C-25) = 6.62 m³

Área de forma = 104.44 m²

CAP FELIPE MIRANDA - IME